



SWA-PF: Semantic-weighted adaptive particle filter for memory-efficient 4-DoF UAV localization in GNSS-denied environments

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ABSTRACT

Vision-based Unmanned Aerial Vehicle (UAV) localization systems have been extensively investigated for Global Navigation Satellite System (GNSS)-denied environments. However, existing retrieval-based approaches face limitations in dataset availability and persistent challenges including suboptimal real-time performance, memory inefficiency for onboard deployment, environmental sensitivity, and limited generalization capability, particularly in dynamic or temporally varying environments. To overcome these limitations, we present a large-scale Multi-Altitude Flight Segments dataset (MAFS) for variable altitude scenarios and propose a novel Semantic-Weighted Adaptive Particle Filter (SWA-PF) method. This approach integrates robust semantic features from both UAV-captured images and satellite imagery through three core innovations: a compact 4-DoF particle representation with strictly linear memory complexity, a semantic weighting mechanism, and an optimized particle filtering architecture. The compact particle design achieves negligible per-particle memory overhead, enabling memory-efficient deployment on resource-constrained UAV onboard computing platforms. Evaluated using our dataset, the proposed method achieves approximately 10× end-to-end computational efficiency gain over classical feature-based methods when embedded in the same particle filter localization framework, maintains global positioning errors below 10 meters with respect to onboard GNSS-based pose measurements, and enables rapid 4 degree of freedom (4-DoF) pose estimation within seconds using accessible low-resolution satellite maps. Code and dataset will be available at <https://github.com/YuanJiayuu/SWA-PF>.

1. Introduction

UAVs are widely deployed across various domains [1]. While the Global Navigation Satellite System (GNSS) is the primary localization solution [2], its vulnerability in complex environments [3] has driven significant interest in autonomous cross-view UAV-to-satellite localization [4]. However, practical large-scale deployments demand continuous flights at variable altitudes with strict onboard computational constraints, necessitating robust, real-time 4-DoF pose estimation.

Current absolute localization methods, including template matching [5,6], feature-based approaches [7,8], visual odometry [9,10], and deep learning [11–13], typically frame localization as image-to-map matching. They often rely on paired data, assume fixed altitudes, and restrict estimation to 2-DoF or 3-DoF. In realistic unconstrained flights, altitude changes and barometer drift inevitably introduce scale variations, while onboard compasses suffer from electromagnetic interference. Consequently, full 4-DoF estimation (x, y, z, θ) is operationally essential to simultaneously resolve both scale and rotational ambiguities.

To address these continuous 4-DoF tracking requirements, particle filtering (PF) is highly suitable as it inherently accommodates non-Gaussian, multi-modal uncertainties without the historical graph overhead of optimization techniques or the scene-specific pre-training of neural implicit models. Yet, conventional PF implementations are computationally expensive and sensitive to appearance variations [14], often demanding memory that scales poorly with the geographic search area. Thus, a critical gap persists: existing methods struggle to simultaneously achieve continuous 4-DoF pose estimation, real-time performance, and strictly linear memory efficiency for resource-constrained onboard deployment.

Furthermore, the rigorous evaluation of such 4-DoF algorithms is fundamentally hindered by the limitations of existing UAV localization datasets [13,15–17]. Most current benchmarks utilize one-to-one paired data, which presents three key limitations: (1) sparse, discontinuous trajectory sampling [17] that prevents the evaluation of temporal filtering; (2) fixed-altitude acquisition that restricts scale variation

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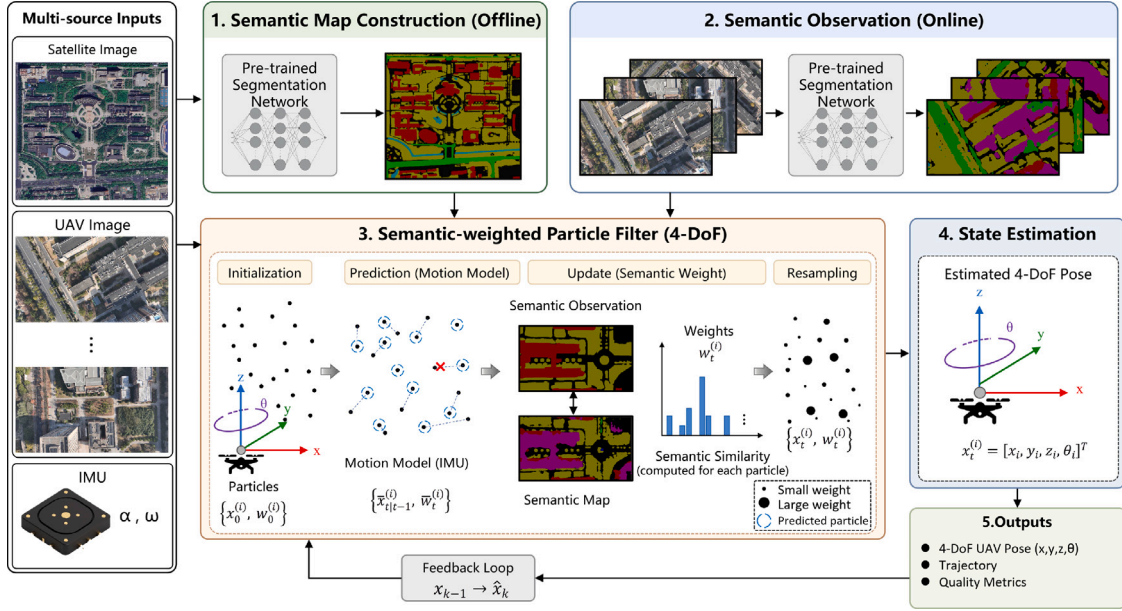


Fig. 1. The simplified pipeline of the proposed visual positioning method SWA-PF. By explicitly modeling the UAV state as $[x_t, y_t, z_t, \theta_t]^T$, the system accounts for altitude and yaw uncertainties. The particle filter recursively fuses IMU-based motion predictions with semantic observation updates, where particle weights are determined by the structural alignment between online UAV semantic masks and the offline global satellite map.

robustness analysis; and (3) rigid paired structures that limit large-area search assessment.

To overcome the fragility of appearance-based matching, semantic scene understanding offers a robust alternative. While raw pixel appearances fluctuate drastically, underlying structural layouts remain consistent. Weighting particles by semantic consistency naturally down-weights ambiguous regions and amplifies discriminative geometric structures, enabling effective matching even with low-resolution satellite maps.

Motivated by these observations, we formulate UAV localization as a full 4-DoF estimation problem ($x_t = [x, y, z, \theta]$) and propose a holistic framework. To address evaluation constraints, we introduce the *Multi-Altitude Flight Segments (MAFS)* dataset and its pixel-level annotated extension, *SemanticMAFS*. Algorithmically, we propose the *Semantic-Weighted Adaptive Particle Filter (SWA-PF)*, whose overall pipeline is depicted in Fig. 1. SWA-PF integrates a semantic-weighted observation model with a strictly linear-memory particle representation. By utilizing a Semantic-Weighted Distance Map (SWDM), we eliminate the need to store and match high-dimensional feature embeddings. The memory requirement per particle is reduced to a minimalist 4-DoF state vector, maintaining a strict global memory complexity of $\mathcal{O}(N)$ and decoupling spatial search resolution from memory consumption.

The main contributions of this work are summarized as follows:

1. We propose SWA-PF, a real-time semantic-aware framework that formulates localization as a 4-DoF estimation problem, overcoming the restricted DoF assumptions of existing approaches.
2. We construct MAFS, a large-scale multi-altitude continuous-trajectory benchmark addressing fixed-altitude and sparse-sampling limitations.
3. We provide SemanticMAFS with pixel-level annotations to enable semantic-aware probabilistic localization research.
4. We release the datasets and source code to promote reproducibility and facilitate future research.
5. We show SWA-PF achieves strong robustness against low-resolution satellite imagery, enabling practical deployment with limited bandwidth and storage.

2. Related works

This section briefly reviews cross-view localization datasets, absolute UAV localization algorithms, and semantic-based localization methods, providing context for the design of our MAFS dataset and SWA-PF framework.

2.1. Comparison of cross-view localization datasets

Existing cross-view localization datasets have transitioned from ground-to-satellite matching under static viewpoints to incorporating aerial perspectives. CVUSA [15] pairs panoramic ground-level images with satellite imagery across large geographic regions, enabling large-scale one-to-one matching. To explicitly model UAV viewpoints, University-1652 [17] introduces tri-view matching by combining ground, UAV, and satellite imagery from 1652 university buildings. SUES-200 [16] investigates altitude variation by capturing UAV images at four discrete flight altitudes. UL14 [13], an extension of DenseUAV [18], introduces dense spatial sampling and multi-temporal satellite imagery, formulating UAV localization as a one-to-many matching task.

Despite this progress, most existing benchmarks fundamentally rely on fragmented sampling strategies or rigid fixed-altitude assumptions. Recent formal benchmarks further expand cross-view evaluation dimensions: The SatLoc dataset [19] provides synchronized multi-source UAV-satellite data for GNSS-denied scenarios, but its altitude range is limited and lacks continuous variable-altitude trajectory annotations. ViLD [20] supports self-supervised cross-view localization with high-altitude UAV imagery, yet it does not provide frame-level IMU and pose metadata for filtering-based sequential evaluation. This inherently restricts their ability to support continuous, variable-altitude localization evaluation under realistic flight conditions, highlighting the necessity of a temporally continuous, multi-altitude benchmark like MAFS.

2.2. Absolute visual localization algorithms for UAVs

Absolute UAV localization methods face distinct trade-offs between robustness, computational cost, and generalization. **Template matching methods** [5,6] estimate pose via direct pixel-level comparison but

Table 1
Summary of representative geo-localization datasets.

Dataset	Altitude (m)	Platform	Target	Sampling	Source	Imgs./Platform
UL14 [13]	90–120	Drone, Satellite	UAV	Discrete	Real Scenes	3 + 6
CVUSA [15]	–	Drone, Ground, Satellite	User	Discrete	Google Maps	1 + 1
SUES-200 [16]	150–300	Drone, Satellite	Diverse	Discrete	Google Maps	50 + 1
University-1652 [17]	–	Drone, Ground, Satellite	Building	Discrete	Google Maps	54 + 16.64 + 1
MAFS (Ours)	100–500	Drone, Satellite	UAV	Dense	Real Scenes	> 100 + 1

are highly sensitive to viewpoint, illumination, and seasonal variations. **Feature-based methods** [7,21] offer improved scale and rotation invariance but incur prohibitive computational costs and degrade in texture-sparse scenes. **Visual odometry** [9,10] tracks inter-frame relative motion but inevitably suffers from long-term global drift. Recently, **deep learning-based methods** [11–13] have significantly improved matching robustness. Liao et al. [22] propose an adaptive threshold-guided ring partitioning framework to improve UAV-satellite matching stability under viewpoint variations. Xiao et al. [23] design a hierarchical absolute localization system that integrates image retrieval and keypoint registration for low-altitude drones. The SatLoc-Fusion framework [19] adopts a hierarchical adaptive fusion strategy to combine visual geo-localization with optical flow tracking, achieving real-time performance on edge devices. However, these approaches typically rely on paired training data, assume fixed flight altitudes, restrict pose estimation to 2-DoF or 3-DoF, and exhibit limited zero-shot generalization to unseen environments. These limitations directly motivate our development of a robust, continuous 4-DoF probabilistic framework.

2.3. Semantic-based scene localization methods

To overcome the fragility of appearance-level matching, recent studies have leveraged high-level semantic scene understanding (e.g., semantic graph matching, histograms, bag-of-words, and Gaussian mixtures) [24–27]. The latest R2Ploc framework [28] further designs a hierarchical semantic aggregation module to fuse multi-scale semantic features for region-to-point UAV geo-localization, achieving improved robustness against scale and viewpoint shifts. However, most existing semantic methods formulate localization as a discrete database matching task.

In contrast, our proposed semantic-aware probabilistic localization explicitly models 4-DoF pose estimation as a continuous Bayesian filtering process. Instead of nearest-neighbor retrieval, SWA-PF utilizes a Semantic-Weighted Distance Map (SWDM) to directly evaluate the observation likelihood of continuous particle states. This inherently preserves multi-modal hypotheses, integrates temporal motion priors, and naturally accommodates continuous variations in altitude and yaw without the overhead of maintaining an exhaustive template database.

3. Dataset

To evaluate the proposed semantic-weighted localization framework, we construct the Multi-Altitude Flight Segments (MAFS) dataset with synchronized UAV imagery, satellite maps, and trajectory annotations, along with its pixel-level semantic extension dataset (SemanticMAFS).

3.1. MAFS dataset

MAFS is a large-scale benchmark for continuous UAV localization. As summarized in Table 1, existing datasets adopt fragmented or sparsely sampled trajectories, whereas MAFS features long temporally continuous flight sequences to support sequential and filtering-based evaluation.

MAFS covers 14 university campuses and a 2 km × 2 km urban region in Hangzhou, China, collected by a DJI Mavic 3E UAV. As shown

in Fig. 2, it includes fixed-altitude and variable-altitude flight modes covering an altitude range of 100–500 m.

The dataset has three core advantages: (1) It supports multi-altitude evaluation with both fixed and continuous variable-altitude trajectories, filling the altitude variation gap in prior benchmarks. (2) It provides tightly synchronized 4K/30fps video and IMU data with frame-level pose metadata (latitude, longitude, altitude, yaw, velocity) for temporal-continuous localization research. (3) It covers diverse urban scenes with a total trajectory length over 40.5 km, each paired with a geo-referenced Google Maps satellite image (zoom level 19).

Survey-grade RTK/PPK ground truth is not included. The UAV integrates a dual-frequency GNSS module (nominal ± 0.1 m RTK, ± 0.5 m standalone accuracy under open sky). All data are collected in open campus environments with minimal signal obstruction, so onboard GNSS measurements are adopted as the error evaluation reference, a widely accepted proxy for GNSS-denied scenarios.

3.2. SemanticMAFS dataset

SemanticMAFS extends MAFS with pixel-level semantic annotations for both views to support semantic cross-view localization. Frames are uniformly sampled and manually annotated via Eiseig [29]: satellite images have 4 categories (building, ground, road, vegetation), and UAV images follow the VDD protocol with 7 categories (roof, wall, ground, road, vegetation, water, vehicle). Representative annotated samples are presented in Fig. 3, and average per-image annotation time exceeds 60 min due to high scene complexity.

Per-class pixel distribution is summarized in Table 2. Vegetation dominates both views (53.81% UAV, 58.37% satellite), while stable structural classes (roofs, roads) provide reliable alignment cues. Inter-annotator agreement exceeds 99% pixel overlap, confirming high annotation consistency. The dataset partitioning for reproducible evaluation is detailed in Table 3, and joint training with VDD mitigates the limited scale of manual annotations.

3.3. Dataset availability

MAFS, SemanticMAFS, source code, pretrained models, and experiment scripts will be publicly released under an academic non-commercial license (v1.0 tag for reproducibility), with long-term maintenance following semantic versioning.

4. Method

This section presents the SWA-PF framework, including state representation, semantic-weighted measurement modeling, and particle filter inference. Inspired by Monte Carlo Localization [14], we adopt particle filter (PF) as the core visual-inertial localization framework, which represents pose distribution with particles and updates via Bayesian inference. To address computational challenges in complex aerial scenes, we design two core components: a multi-altitude semantic segmentation module and a semantic-weighted adaptive PF localizer, as outlined in Fig. 4.

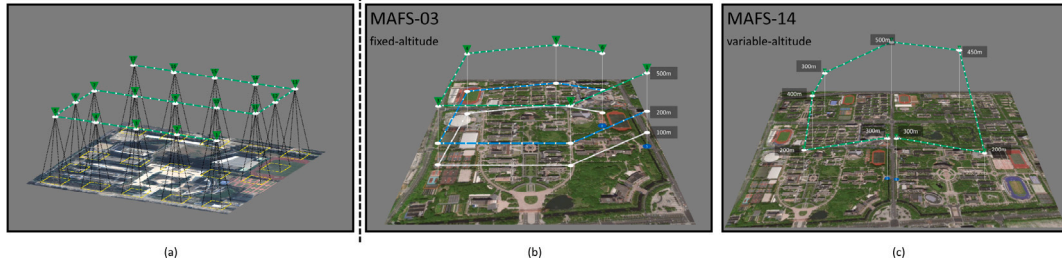


Fig. 2. Comparison of dataset flight strategies. (a) Conventional retrieval-based datasets. (b) Fixed-altitude flights at 100–500 m. (c) Variable-altitude flights ranging from 150–500 m.

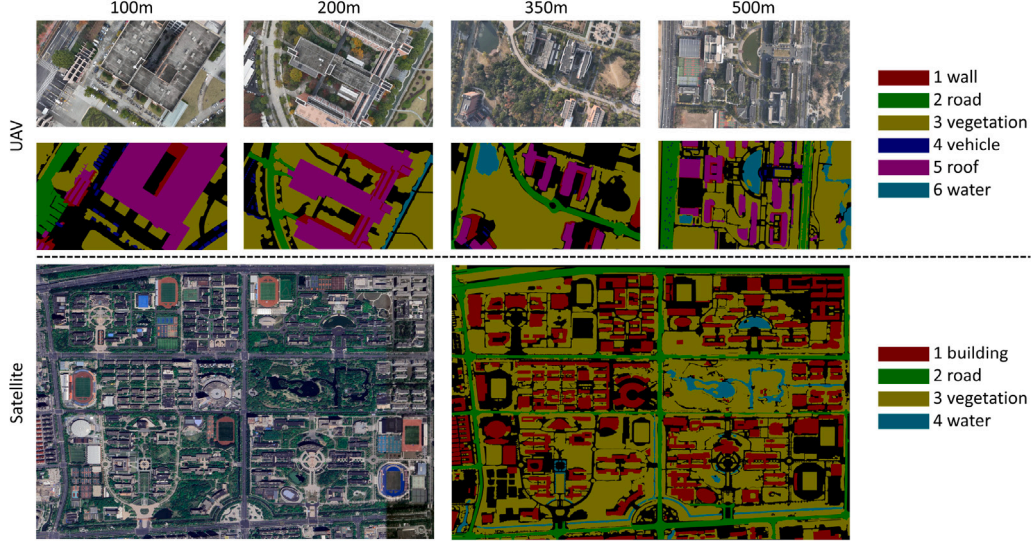


Fig. 3. Representative examples from SemanticMAFS. The top row shows UAV top-view images with seven semantic categories, while the bottom row shows corresponding satellite images with four categories.

Table 2
Per-class pixel distribution of SemanticMAFS.

	class	walls	roof	road	vegetation	vehicles	water
UAV	Pixel Count	1.86e+07	9.52e+07	2.03e+07	1.68e+08	3.84e+06	6.74e+06
	Percentage (%)	5.95	30.38	6.48	53.81	1.22	2.14
	class	building	road	vegetation	–	–	water
Satellite	Pixel Count	1.92e+07	1.17e+07	4.68e+07	–	–	2.39e+06
	Percentage (%)	23.98	14.63	58.37	–	–	2.99

Table 3
Dataset partition statistics of SemanticMAFS.

Task	Dataset	Train	Val
Satellite Segmentation	SemanticMAFS	11 images	4 image
	SemanticMAFS + VDD	76 images + 320 images	13 images + 80 images

4.1. Semantic segmentation

We perform semantic segmentation on both satellite and UAV images. For satellite imagery, a VGG-pretrained U-Net is trained on three densely annotated Google Maps images, which suffices to capture the semantic structures required for localization given consistent visual styles.

UAV top-view images are more challenging due to altitude variation and partial building visibility. We adopt a 7-class scheme following the

VDD protocol [30] and select SegFormer-B0 as the backbone to balance accuracy and onboard efficiency.

As shown in Fig. 5, despite real-world noise, segmentation quality suffices for downstream localization. Stochastic data augmentation is applied during training to improve generalization.

4.2. Semantic-weighted adaptive particle filter

To estimate the 4-DoF UAV pose from 2D satellite imagery, we extend the localization problem into 3D space using a particle filter. Each particle represents a UAV state $x_t = [x, y, z, \theta]$, where (x, y, z) denote the 3D position (with z representing altitude) and θ denotes the yaw angle.

4.2.1. Prediction

The system continuously fuses the control input u_t , containing linear velocity v_t and angular velocity ω_t from odometry, with the semantic observation z_t from the downward-facing camera. Based on standard

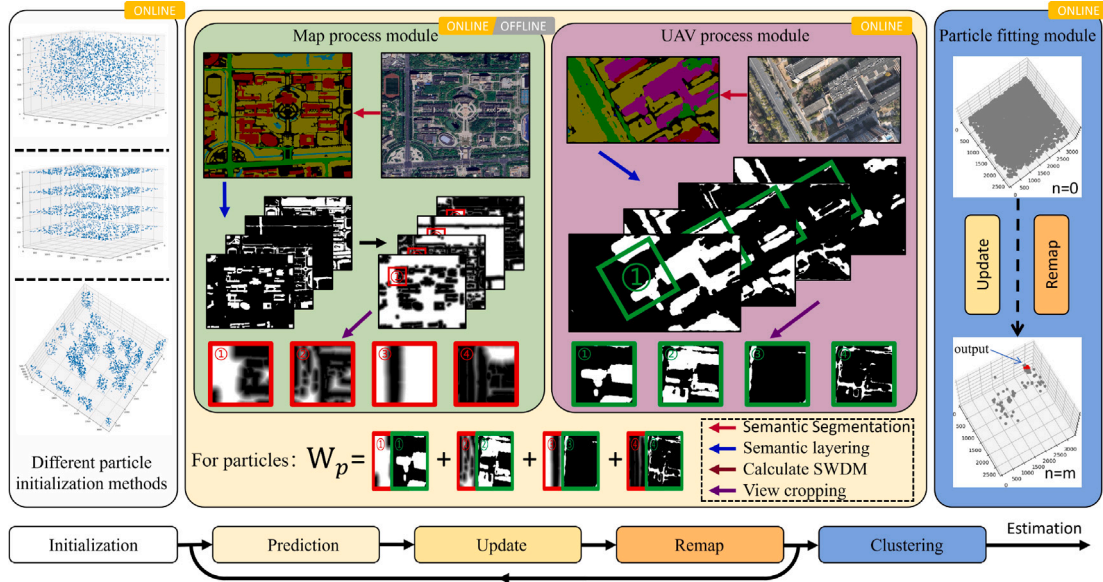


Fig. 4. Overall framework of the proposed SWA-PF. Particle states are initialized with prior knowledge. During prediction, real-time semantic segmentation of UAV top-view images is used to compute particle weights. The UAV position is then estimated through iterative updating and clustering of particles.

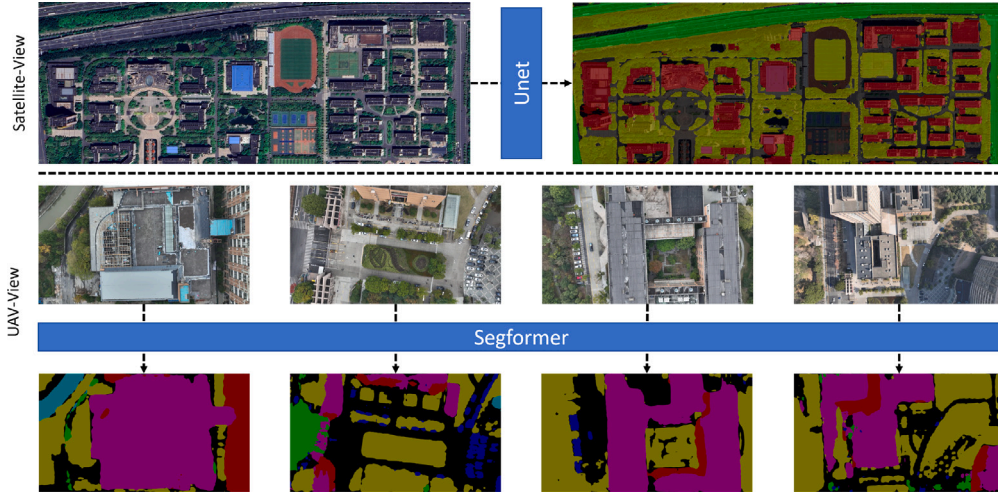


Fig. 5. Semantic segmentation results for satellite and UAV images using U-Net [31] and SegFormer [32]. Both modalities exhibit real-world noise.

Bayesian filtering, the posterior probability distribution of the state is recursively estimated via prediction and update steps:

$$P(x_t | z_{1:t-1}, u_{1:t}) = \int P(x_t | x_{t-1}, u_t) P(x_{t-1} | z_{1:t-1}, u_{1:t-1}) dx, \quad (1)$$

$$P(x_t | z_{1:t-1}, u_{1:t}) = \frac{P(z_t | x_t) P(x_t | z_{1:t-1}, u_{1:t-1})}{P(z_t | z_{1:t-1}, u_{1:t-1})}. \quad (2)$$

Specifically, the state transition model updates each particle according to the following kinematic model:

$$P(x_t | x_{t-1}, u_t) = x_{t-1} + v_{t-1} \Delta t + \omega_{t-1} \Delta t + \epsilon, \quad (3)$$

where the state variables x, y, z , and θ are corrupted by an additive Gaussian noise vector $\epsilon \sim \mathcal{N}(0, \sigma^2)$. This noise component explicitly simulates inherent IMU drift and dynamic perturbations during evaluation.

4.2.2. Measurement model

In the measurement model, the semantic observation z_t encapsulates pixel coordinates and structural labels. To achieve scale invariance and mitigate significant field-of-view variations caused by altitude

changes z (as shown in Fig. 6), both the UAV-view image z_t and the locally cropped semantic map M are normalized to a fixed dimension of 400×400 pixels.

Rather than calculating fragile pixel-wise semantic consistency, we compute the observation likelihood using a Semantic-Weighted Distance Map (SWDM) $M_{(\text{nearest}|p)}$:

$$M_{(\text{nearest}|p)} = \sum_{L=0}^H \sum_{j=0}^W \min_{L_{ij}=L} \|p_{ij} - p\|_2. \quad (4)$$

This metric calculates the Euclidean distance to the nearest semantically equivalent point across all pixels in the image. To provide a concrete visual example, Fig. 7 illustrates the Semantic Weighted Distance Map for the MAFS-10 path within satellite imagery. Additional high-resolution semantic segmentation masks and SWDM visualizations are provided in Appendix.

To enhance vertical convergence and emphasize central semantic similarities, we incorporate a center distance field matrix M_{CDF} (as shown in Fig. 8). The weight W for each particle is then computed as:

$$W = \sum_L \sum_p \frac{\alpha_L}{z_t \cdot M_{(\text{nearest}|p)}} \cdot M_{CDF} + \gamma. \quad (5)$$

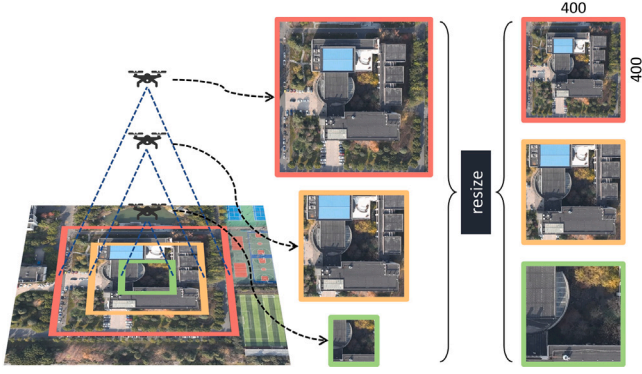


Fig. 6. Illustration of altitude effects on particle viewpoints. Particles at different altitudes yield distinct fields of view, influencing weight calculation. To address this, all particle viewpoints are resized to 400×400 pixels.

where α_L denotes the weight coefficient corresponding to the L -th semantic category, which adjusts the contribution of different semantic structures to the particle weight update.

4.2.3. Particle representation and memory efficiency

A key advantage of SWA-PF is its compact particle representation: each particle stores only a 4-DoF state vector, forming a lightweight $N \times 4$ tensor with strictly linear $\mathcal{O}(N)$ memory complexity. This design yields negligible per-particle overhead, far lower than feature-matching methods that maintain high-dimensional descriptors. It enables high-density PF on SWaP-constrained onboard platforms without memory exhaustion. As verified in experiments, GPU memory is dominated by semantic map tensors rather than Semantic observation of particles, ensuring strong scalability for large-area localization.

4.2.4. Parameter setting and tuning

Motion noise is set to $\epsilon = 15$ (Eq. (3)), exceeding typical IMU error to emulate challenging conditions. Semantic weights α_L mitigate class imbalance in the measurement model, rebalancing class contributions and improving sensitivity to geometrically meaningful regions. The normalization constant $\gamma = 10$ balances convergence speed and prevents premature particle collapse.

Notably, no trajectory-specific tuning is performed: a single unified hyperparameter set is used across all trajectories, datasets, and ablation studies to ensure fair comparison and validate framework robustness.

4.3. Performance optimization

Discretized Yaw Optimization. Image rotation is a major PF computational bottleneck. We discretize yaw into 100 uniform bins (3.6° per bin) and pre-rotate UAV images to these angles. As demonstrated in Fig. 9, particles then index the nearest pre-computed rotation, reducing per-particle rotation overhead from $\mathcal{O}(N)$ to a constant $\mathcal{O}(100)$ per iteration with negligible quantization error.

Matrix Batch Optimization. Further acceleration is achieved via optimized matrix stacking, eliminating explicit loops in weight calculation. With GPU parallelization, 20,000 particles are evaluated in 4.9 s.

Density-Based State Estimation. Direct averaging of posterior particles causes bias from outliers. We use DBSCAN [33] to remove low-density outliers and identify dominant clusters, then compute cluster centroids as the final estimate once posterior covariance falls below a threshold, yielding robust estimation under noisy conditions.

4.4. Comparison with classical particle filter pipeline

Fig. 10 compares the classical PF pipeline with SWA-PF. A conventional PF pipeline includes particle initialization, motion prediction, feature-based observation computation, weight normalization and resampling, and weighted-average state estimation.

SWA-PF introduces four core improvements: (1) semantic-aware measurement model based on SWDM; (2) center distance field weighting for enhanced convergence stability; (3) discretized yaw optimization for reduced rotational cost; (4) DBSCAN-based robust state estimation. These upgrades transform classical PF into a semantic-weighted, memory-efficient 4-DoF localization framework. Detailed pseudocode of the proposed algorithm is provided in Appendix.

5. Experiments

To validate the effectiveness and efficiency of SWA-PF, we conduct extensive experiments on MAFS, comparing it with traditional and learning-based methods under fixed- and variable-altitude settings. The evaluation is systematically structured as follows: Section 5.1 establishes the experimental setup and benchmarks the core performance, transitioning from computational efficiency analysis to comprehensive localization accuracy assessment. Subsequently, Section 5.2 presents detailed ablation studies to isolate the impact of specific algorithmic components, followed by a comparison with existing state-of-the-art frameworks in Section 5.3.

5.1. Comparison of particle filter-based localization algorithms

To comprehensively evaluate the proposed SWA-PF, we established a rigorous experimental setup encompassing diverse flight scenarios, multiple baseline comparisons, and a systematic set of evaluation metrics.

5.1.1. Datasets and baselines

SWA-PF is evaluated on three MAFS trajectories: two fixed-altitude (200 m) scenarios (MAFS-03, MAFS-10) and one variable-altitude (150–500 m) scenario (MAFS-14). We use 5000 particles for MAFS-03/10 and 40,000 for MAFS-14 to accommodate the larger search space. Particles are uniformly sampled based on the dataset protocol: $x \in [0, W_{map}]$, $y \in [0, H_{map}]$, $z \in [50, 500]$ m.

We compare SWA-PF with traditional feature-based and template-matching methods (MI, HIST, SIFT, AKAZE, BRISK, ORB) and deep learning-based baselines (OS-FPI, DenseUAV, SSPT). All traditional methods are integrated into the same particle filter framework as SWA-PF to ensure fairness.

The motion noise is set to $\epsilon = 15$ to emulate challenging conditions, and the normalization constant is fixed at $\gamma = 10$. To balance statistical reliability and computational cost, all particle filter-based methods share identical random seeds across 20 independent runs for the same run index, ensuring identical particle initialization and stochastic noise realization. The adequacy of this sample size ($n = 20$) is mathematically supported by post-hoc statistical power analysis. Because statistical power is fundamentally driven by the magnitude of the effect size, the exceptionally large performance gaps observed in our experiments (with Cliff's δ consistently exceeding the threshold for a “Large” effect, $|\delta| \geq 0.474$) ensure that $n = 20$ yields a statistical power ($1 - \beta$) well above the conventional 0.80 threshold. This confirms that the evaluation is adequately powered to detect true performance differences without a significant risk of Type II errors.

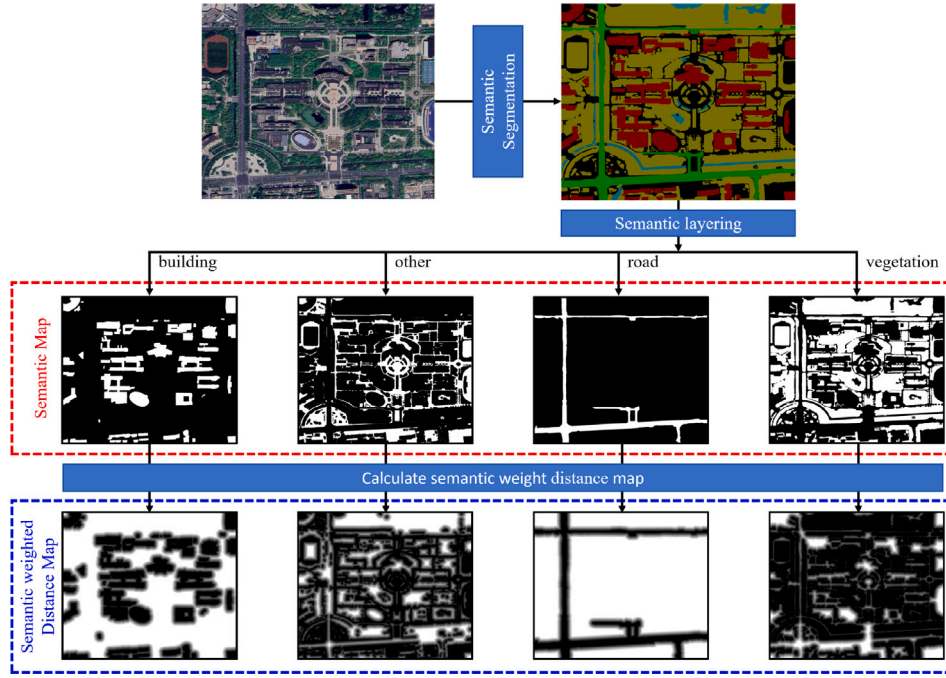


Fig. 7. Classification semantic map and the Semantic Weighted Distance Map (SWDM). After semantic segmentation of satellite imagery, the SWDM is computed for each layer, either during preprocessing or real-time processing.

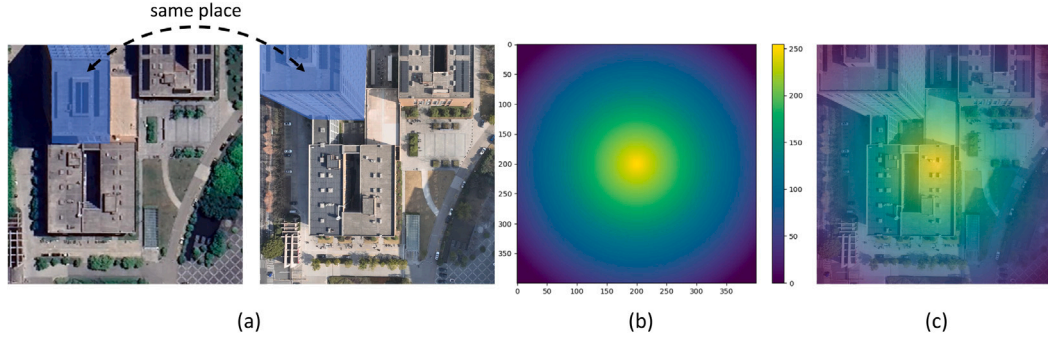


Fig. 8. (a) An example demonstrates that even the same building exhibits significant differences in style and perspective between UAV and satellite image. (b) Center distance field matrix M_{CDF} . (c) We use M_{CDF} to increase the center weight and encourage center alignment.

5.1.2. Evaluation metrics

We systematically assess performance along three dimensions – computational efficiency, localization accuracy, and statistical reliability – as summarized in Table 4. Computational efficiency metrics are Fitting Time (s), the duration from particle initialization to convergence (covering semantic segmentation, state prediction, weight computation, resampling, and final state estimation), and Finish Time (s), the total cumulative time needed to complete full trajectory localization from the initial frame. Localization accuracy is measured by RMSE (p) (root mean square error of global position estimates), Median (p) (median error, robust to extreme outliers), and Recall@10 (%) (percentage of frames with position error < 10 m). For statistical evaluation (vs. SWA-PF), we report means and 95% confidence intervals (CI) over 20 independent runs, using the format $a \pm b$. Because absolute positioning errors are zero-bounded and severely right-skewed with occasional extreme outliers, we employ pairwise Wilcoxon signed-rank tests with Holm-Bonferroni correction ($\alpha = 0.05$) instead of parametric alternatives. The p -value tests the null hypothesis of no median RMSE difference, while Cliff's δ quantifies effect size (negligible $|\delta| < 0.147$, small $0.147 \leq |\delta| < 0.33$, medium $0.33 \leq |\delta| < 0.474$, large $|\delta| \geq 0.474$).

5.1.3. Computational efficiency

Having defined the timing metrics, we evaluate the real-world inference efficiency of various image matching methods using the MAFS-10 benchmark. All experiments, including map preprocessing, semantic segmentation, and real-time localization, are conducted on a simulation platform equipped with an Intel Core i5-12400F CPU and an NVIDIA GeForce RTX 4060Ti GPU.

To maximize hardware utilization, all implementations employ multi-threaded processing. As detailed in Table 4, despite the inherent computational complexity associated with semantic feature extraction, SWA-PF demonstrates a substantial speed advantage. To clarify the operational pipeline and the exact scope of our runtime measurements: in our experimental setup, only the semantic segmentation of the global satellite reference map is precomputed offline prior to flight to minimize initialization overhead. During the continuous localization process, the semantic segmentation inference for the incoming UAV top-view images, the dynamic generation of the Semantic-Weighted Distance Maps (SWDM), and all subsequent particle filtering updates operate fully online in real-time. Consequently, the reported runtime metrics (both Fitting Time and Finish Time) strictly encompass all these online computational overheads. Thanks to optimized GPU acceleration

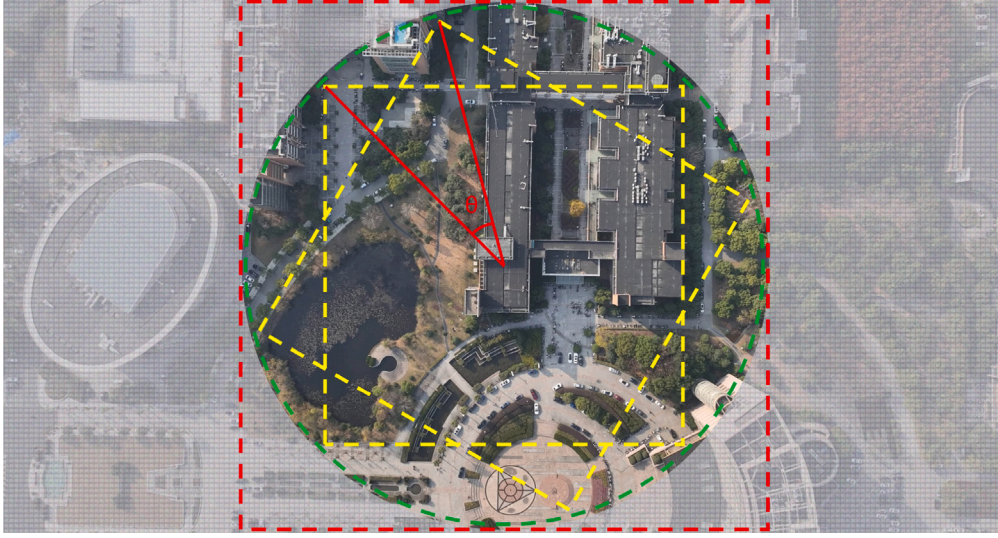


Fig. 9. Rotation optimization during particle weighting. The particle direction angle is matched to the nearest discrete angle. The yellow square denotes the preserved rotated image, while the gray region indicates discarded areas. The step size of θ is 3.6° .

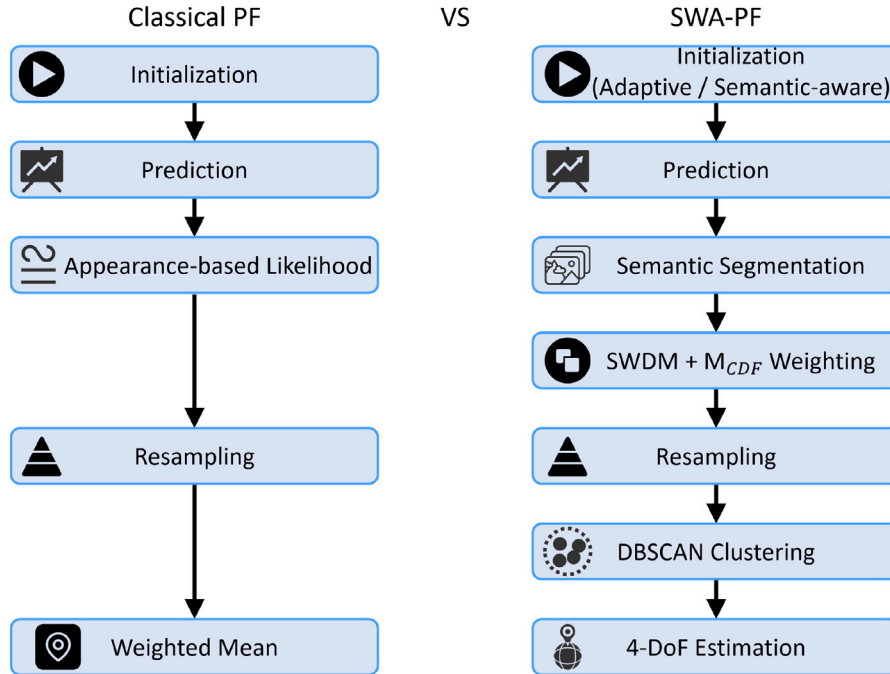


Fig. 10. Simplified comparison between a classical particle filter localization pipeline and the proposed SWA-PF framework.

combined with our multi-threaded architecture, our method achieves superior Fitting Time and Finish Time compared to traditional high-dimensional feature-based approaches. These efficiency gains confirm that the proposed framework satisfies the strict real-time processing constraints necessary for practical deployment on medium-sized UAV platforms.

To further characterize the full distribution of localization errors and evaluate the stability of each method across 20 independent runs, we present a compact violin plot of global localization errors for all compared methods in Fig. 11.

Given the significant performance gaps and the presence of extreme outliers in traditional feature-based methods (e.g., SIFT, BRISK) and learning-based methods (e.g., OS-FPI, DenseUAV), we explicitly utilize a logarithmic scale for the y-axis. This logarithmic violin plot not only clearly illustrates the probability density of the errors but also perfectly accommodates the massive magnitude differences among

algorithms without visual compression. As illustrated, the proposed SWA-PF achieves the lowest median error and the most concentrated distribution (narrowest interquartile range) among all methods. This indicates not only superior localization accuracy but also minimal performance fluctuation across repeated trials, further validating the robustness of SWA-PF for practical UAV deployment.

Having established both the computational feasibility and the overall error stability of SWA-PF, the subsequent evaluation shifts focus to a detailed assessment of absolute localization accuracy and temporally continuous tracking performance.

5.1.4. Localization accuracy

Fig. 12 compares the minimum global localization error of all methods on the fixed-altitude MAFS-10 dataset. Several baseline methods produce ambiguous pose estimates. Among traditional approaches, AKAZE and HIST achieve relatively high accuracy but incur prohibitive

Table 4

Computational Efficiency and Localization Performance Comparison. All values are reported as mean $\pm$ 95% CI over 20 independent runs. The Interquartile Range (IQR) is newly provided to explicitly quantify error dispersion. The performance metrics of the proposed SWA-PF are highlighted in bold.

Method	Fitting time (s)	Finish time (s)	RMSE (p)	Recall@10 (%)	Median (p)	IQR (p)	p -value (vs. SWA-PF)	Cliff's δ (vs. SWA-PF)
SIFT [34]	1011.06 $\pm$ 159.86	4879.26 $\pm$ 771.48	171.47 $\pm$ 2.85	10.06 $\pm$ 1.59	115.12 $\pm$ 5.64	150.67	< 2.98e-15	0.98 (Large)
MI [35]	290.95 $\pm$ 48.49	1356.97 $\pm$ 226.16	145.80 $\pm$ 2.54	37.82 $\pm$ 7.97	19.58 $\pm$ 0.47	13.33	< 0.007	0.52 (Large)
HIST [36]	104.76 $\pm$ 22.34	583.50 $\pm$ 124.40	62.75 $\pm$ 1.43	44.68 $\pm$ 13.05	18.50 $\pm$ 1.14	16.78	< 0.009	0.49 (Large)
AKAZE [37]	150.39 $\pm$ 23.78	1148.32 $\pm$ 181.57	47.07 $\pm$ 0.75	55.93 $\pm$ 8.84	34.85 $\pm$ 1.61	35.33	< 4.46e-07	0.81 (Large)
BRISK [38]	605.13 $\pm$ 95.68	2959.13 $\pm$ 467.88	78.20 $\pm$ 1.39	25.21 $\pm$ 3.99	49.03 $\pm$ 1.37	56.77	< 1.69e-13	0.95 (Large)
ORB [39]	34.59 $\pm$ 5.47	450.38 $\pm$ 71.21	77.68 $\pm$ 1.24	20.33 $\pm$ 3.21	63.99 $\pm$ 3.12	58.14	< 1.25e-14	0.96 (Large)
SSPT [40]	–	–	225.89 $\pm$ 2.01	45.28 $\pm$ 4.13	59.03 $\pm$ 5.68	299.73	< 7.24e-16	0.97 (Large)
DenseUAV [18]	–	–	192.98 $\pm$ 3.11	3.73 $\pm$ 0.34	108.00 $\pm$ 1.99	139.43	< 3.89e-18	1.00 (Large)
OS-FPI [11]	–	–	261.75 $\pm$ 2.46	31.78 $\pm$ 2.90	108.00 $\pm$ 1.99	336.72	< 1.50e-15	0.98 (Large)
SWA-PF (Ours)	6.13 $\pm$ 0.79	22.32 $\pm$ 2.88	24.37 $\pm$ 0.32	95.76 $\pm$ 12.36	18.91 $\pm$ 0.38	16.25	–	–

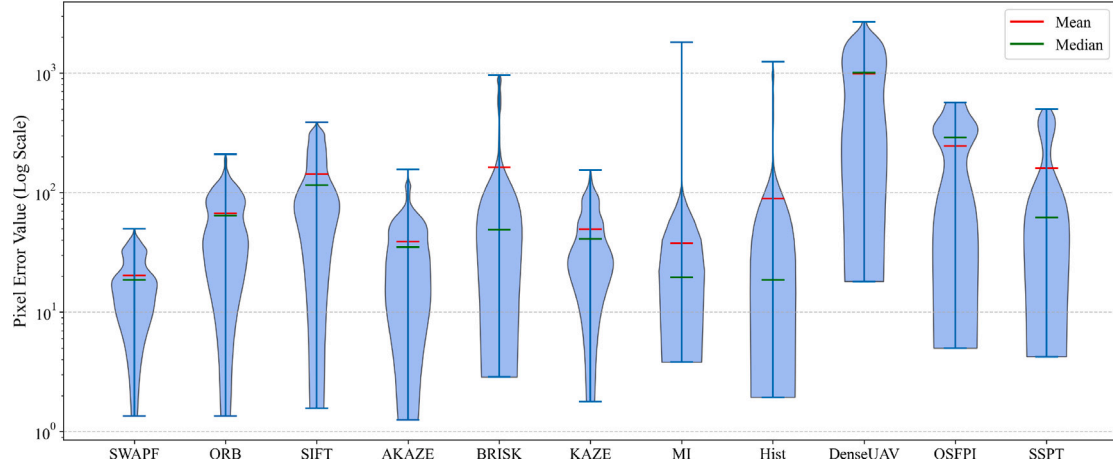


Fig. 11. Violin plot of global localization error distributions for all evaluated methods on the MAFS-10 dataset. The y-axis explicitly utilizes a logarithmic scale to effectively accommodate the massive error gaps and extreme outliers of traditional baseline methods. The red lines indicate the mean, and the green lines indicate the median.

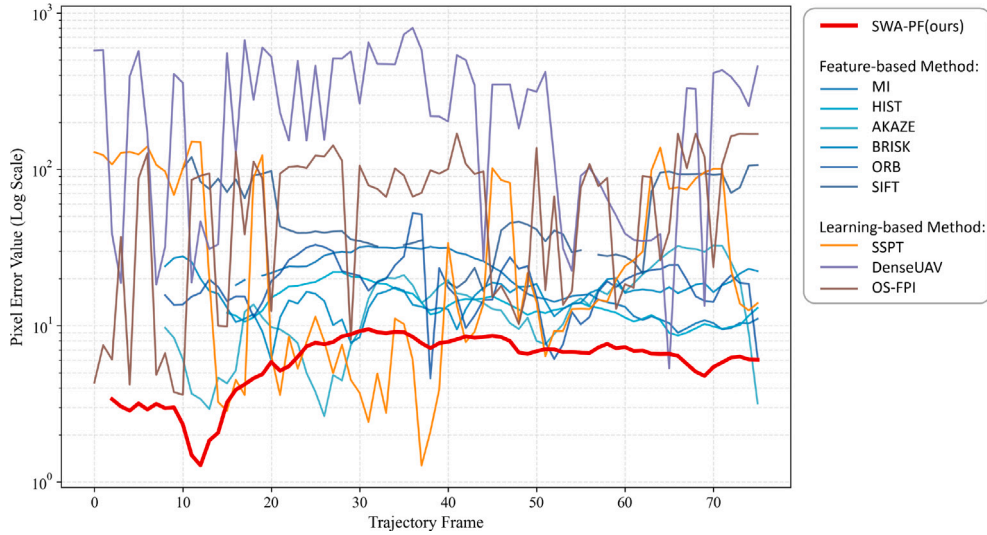


Fig. 12. Lowest global localization error achieved by each evaluated method on the MAFS-10 dataset at a fixed altitude of 200 m.

computational cost, preventing real-time operation. Learning-based methods such as SSPT, DenseUAV, and OS-FPI process frames independently and therefore exhibit discontinuous errors and temporal instability on continuous video sequences.

In contrast, the proposed semantic-based measurement model delivers more accurate and temporally stable localization with improved efficiency, converging within 9 s and processing the entire trajectory in 31 s. Statistical analysis confirms that the performance improvement of SWA-PF over all baseline methods is not only statistically significant

(all $p < 0.05$, Wilcoxon signed-rank test) but also practically meaningful, with Cliff's delta effect sizes indicating large effects ($|\delta| \geq 0.474$) for all pairwise comparisons. This demonstrates that SWA-PF delivers substantial performance gains relative to state-of-the-art baselines.

5.1.5. Evaluation of 4-DoF estimation: Altitude and yaw

Particle convergence becomes significantly more challenging in the variable-altitude MAFS-14 scenario. As expected, conventional methods

Table 5

Performance evaluation of SWA-PF on CPU and GPU across MAFS-03, MAFS-10, and MAFS-14 trajectories. All values are reported as mean $\pm$ 95% CI over 20 independent runs.

Trajectory	CPU/GPU	Fitting time (s)	Finish time (s)	Avg. time/frame (s)	RMSE (p)	Recall@10 (%)	Median (p)
MAFS-03	CPU	102.44 $\pm$ 16.61	267.82 $\pm$ 43.44	3.23 $\pm$ 0.52	32.86 $\pm$ 0.72	74.42 $\pm$ 12.23	22.70 $\pm$ 0.75
	GPU	7.34 $\pm$ 1.16	19.49 $\pm$ 3.08	0.23 $\pm$ 0.04	31.72 $\pm$ 0.74	81.99 $\pm$ 11.38	24.63 $\pm$ 1.23
MAFS-10	CPU	96.31 $\pm$ 15.22	329.87 $\pm$ 52.15	4.34 $\pm$ 0.69	24.43 $\pm$ 0.39	93.29 $\pm$ 14.75	19.16 $\pm$ 0.46
	GPU	6.13 $\pm$ 0.79	22.32 $\pm$ 2.88	0.29 $\pm$ 0.04	24.37 $\pm$ 0.32	95.76 $\pm$ 12.36	18.91 $\pm$ 0.38
MAFS-14	CPU	816.02 $\pm$ 148.58	2760.60 $\pm$ 504.01	43.13 $\pm$ 7.88	38.59 $\pm$ 1.00	69.09 $\pm$ 12.61	32.10 $\pm$ 0.87
	GPU	57.82 $\pm$ 13.56	170.47 $\pm$ 49.21	2.66 $\pm$ 0.77	37.18 $\pm$ 1.64	67.22 $\pm$ 19.40	34.15 $\pm$ 2.41

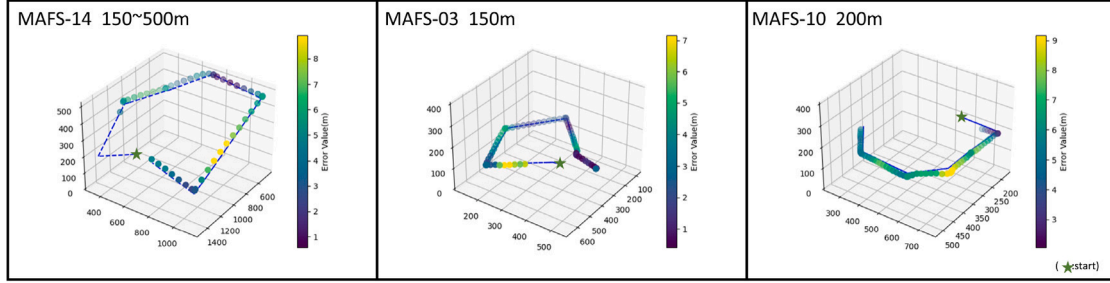


Fig. 13. 3D trajectory results of SWA-PF on MAFS-03, MAFS-10 (fixed altitude) and MAFS-14 (variable altitude). Color bars indicate localization error (m), and the red star denotes the trajectory start point.

almost entirely fail, whereas the proposed 3D SWA-PF remains effective despite substantial altitude variations. Although 3D localization slightly slows convergence, the required number of iterations remains comparable to that in fixed-altitude cases. This robustness stems from the proposed weighting function, which maintains high sensitivity to altitude while preserving computational efficiency. To explicitly demonstrate this efficiency and localization accuracy across different hardware configurations, Table 5 details the performance metrics of the SWA-PF method (including the average inference time per frame) using both CPU and GPU implementations across the MAFS-03, MAFS-10, and MAFS-14 trajectories. The robustness of this approach is explicitly visualized in Fig. 13, which presents the resulting 3D trajectories for MAFS-03, MAFS-10, and MAFS-14.

Fig. 14 compares our method's altitude and yaw angle estimates against GNSS-based ground truth across two distinct operational environments: a fixed-altitude flight at 200 m (MAFS-10) and a variable-altitude flight ranging from 150 m to 500 m (MAFS-14). Yaw angle error gradually approaches low or even zero error with displacement, demonstrating our method's superior altitude awareness. However, altitude estimation exhibits some lag, particularly in the variable-altitude MAFS-14 path. We attribute this to motion model latency: errors increase during rapid ascent or descent but are quickly corrected upon receiving new semantic observations. These results confirm that SWA-PF reliably estimates all four degrees of freedom, achieving competitive accuracy not only in horizontal positioning but also in altitude tracking and yaw orientation, thereby fulfilling the core requirements of the proposed framework.

5.1.6. GPU memory usage and scalability

GPU memory usage is profiled to evaluate scalability. During initialization, the method consumes approximately 3004.21 MB of GPU memory, increasing to a peak of 3499.58 MB during particle iterations. This increase is dominated by tensor representations of satellite and UAV semantic maps rather than particle storage. Particle states are stored as a compact $n \times 4$ tensor, resulting in negligible per-particle memory overhead. Quantitatively, assuming standard 32-bit floating-point precision, each particle requires merely 16 bytes of memory. Even for the most complex variable-altitude scenario (MAFS-14) utilizing $N = 40,000$ particles, the total state memory footprint is strictly bounded to approximately 0.61 MB. Extrapolating to larger-scale deployments, maintaining 100,000 particles would consume only

about 1.53 MB. This is orders of magnitude lower than the memory required to cache dense visual descriptors or high-dimensional network embeddings in retrieval-based methods. Consequently, the scalability of SWA-PF is primarily constrained by satellite map resolution and coverage, rather than by the number of particles.

In terms of runtime scaling, the per-iteration computational complexity grows approximately linearly with the number of particles $\mathcal{O}(N)$, as each particle requires semantic cropping and weight computation. Increasing the satellite map size beyond the current $2 \text{ km} \times 2 \text{ km}$ setting primarily enlarges the spatial search domain, which may require a higher particle count to maintain spatial coverage density. However, the per-particle state representation remains a compact $n \times 4$ tensor, resulting in negligible memory growth per particle. Therefore, runtime increases are mainly driven by particle density requirements rather than by map size itself, while GPU memory usage is dominated by the semantic map tensor resolution.

5.2. Ablation study

While the comparative evaluations in Section 5.1 demonstrate the overall superiority of SWA-PF against baseline methods, this subsection isolates the internal mechanisms driving this performance. Through targeted ablation experiments, we systematically analyze the sources of positioning errors, the framework's robustness to map resolution degradation, and the specific performance gains contributed by different particle initialization strategies and the center distance field (M_{CDF}).

5.2.1. Positioning error analysis

Fig. 15 presents our systematic analysis of localization errors, which reveals consistent spatial error patterns across repeated path trials. The data indicates significant mid-trajectory positioning inaccuracies, with errors clustered in specific spatial regions. Two primary factors contribute to these observations: (1) Perspective disparity between oblique satellite imagery and UAV top-view images, leading to substantial geometric mismatches that amplify positioning errors. (2) Limited distinctive visual features along the test path, reducing the availability of reference points for spatial alignment. While our method is unable to accurately estimate pose in feature-scarce scenarios (e.g., when large portions of the field of view are dominated by buildings or vegetation), this limitation is common to many existing visual localization methods that struggle in the same single scene environments.

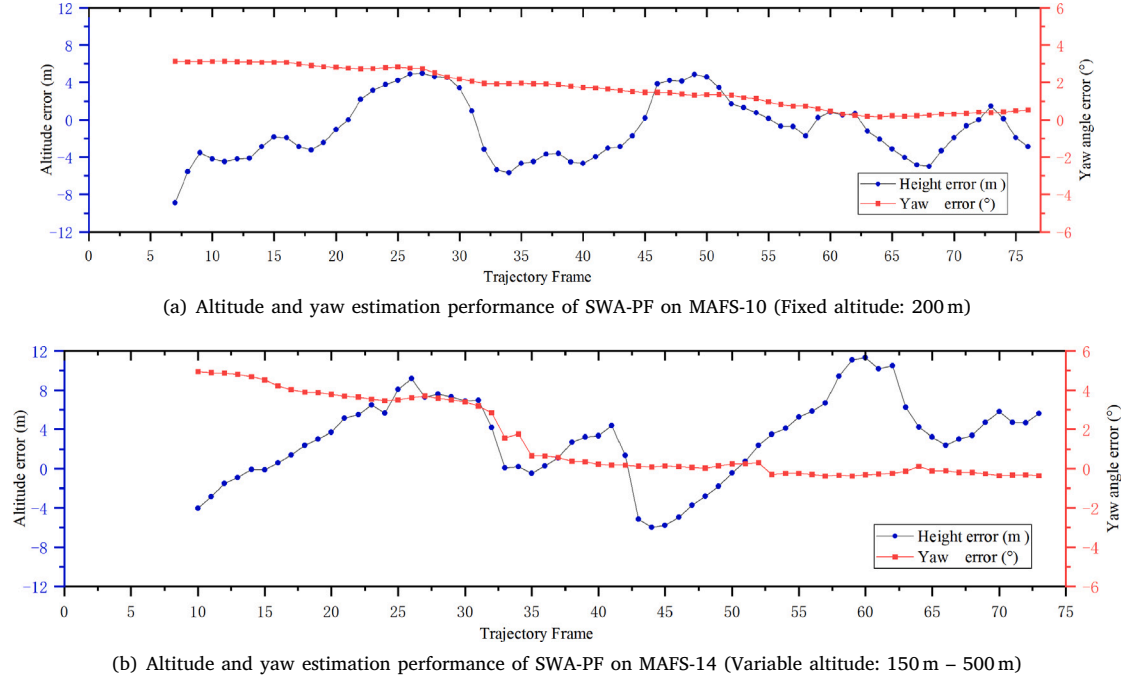


Fig. 14. Altitude and yaw estimation on (a) fixed-altitude (MAFS-10) and (b) variable-altitude (MAFS-14) trajectories.

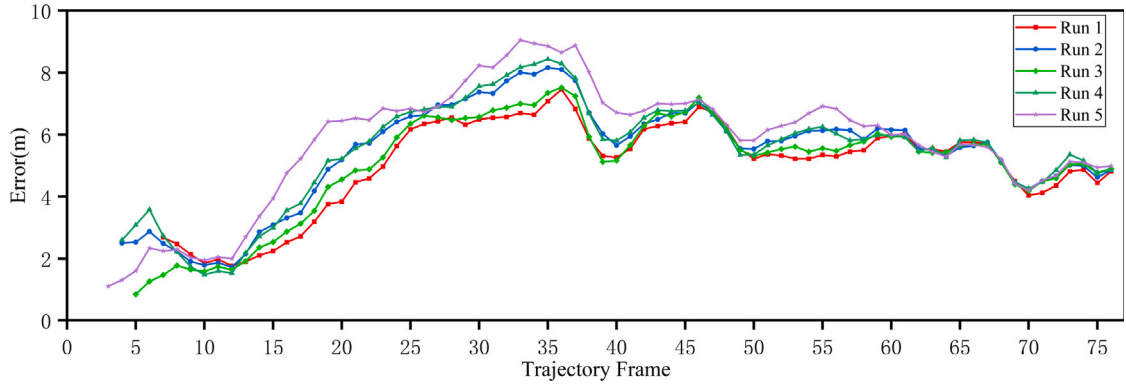


Fig. 15. The lines 1–5 represent the error results from five experiments using the SWA-PF method on the same trajectory. The overall errors exhibit a consistent trend.

5.2.2. Performance under low resolution and temporal map variations

A practical advantage of the proposed framework is its ability to operate reliably with low-resolution satellite imagery. In real-world deployment scenarios, high-resolution maps are not always available due to bandwidth limitations, storage constraints, or outdated satellite coverage.

To evaluate robustness under degraded map quality, we reduce the satellite image resolution by half and re-run the localization experiments. Despite the doubled pixel-to-meter ratio, the resulting localization error exhibits only marginal variation compared with the original-resolution setting, as illustrated in Fig. 16. This behavior indicates that the semantic-weighted measurement model relies primarily on structurally stable semantic distributions rather than fine-grained texture details.

Unlike appearance-driven or dense feature-matching methods, which often degrade significantly under resolution reduction, SWA-PF maintains stable convergence behavior and comparable positioning accuracy. This robustness provides explicit operational advantages for real-world UAV deployment. First, it drastically reduces onboard solid-state storage requirements, allowing resource-constrained edge devices

to cache reference maps for vastly larger geographic areas. Second, in emergency response or remote exploration missions where high-speed communication infrastructure is compromised or unavailable, SWA-PF facilitates on-the-fly map updates via low-bandwidth telemetry links, as downloading gigabytes of high-resolution imagery in such environments is physically infeasible. This robustness substantially enhances practical deployability, allowing localization over large geographic regions using highly accessible, lightweight satellite maps.

5.2.3. Impact of particle initialization methods on convergence rate

Three particle initialization strategies, designed based on semantic cues and prior geographic knowledge, are compared in Fig. 17. Method (a) performs full-space randomization by uniformly distributing particles across the 3D search space. Method (b) adopts layered randomization, where particles are initialized within a single layer and replicated across multiple altitude layers. Method (c) aligns particle clusters with dominant semantic regions in the UAV's central field of view. Experimental results show that method (c) achieves faster initial convergence when salient semantic structures are present. Although method (b) reduces initialization cost, mismatches between preset layer

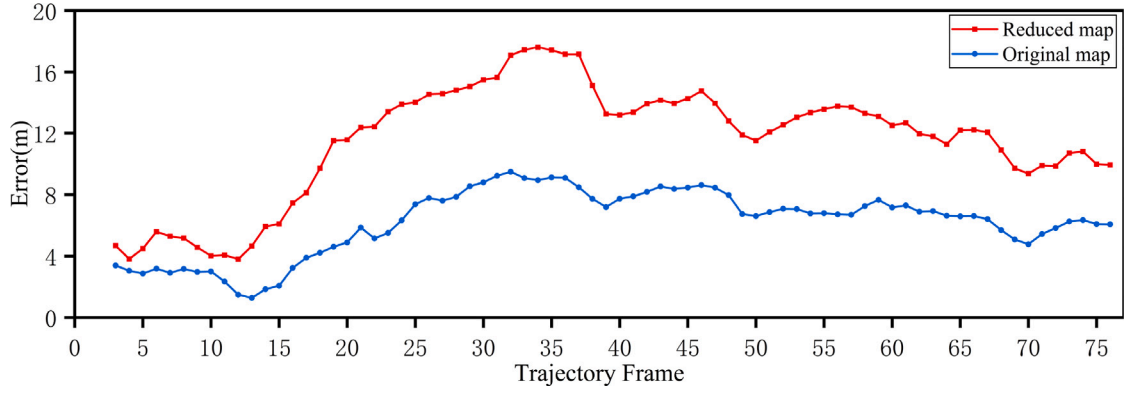


Fig. 16. Comparison of SWA-PF method error in original and low-resolution images. The pixel size of the map was reduced by half, leading to a doubling of the pixel error.

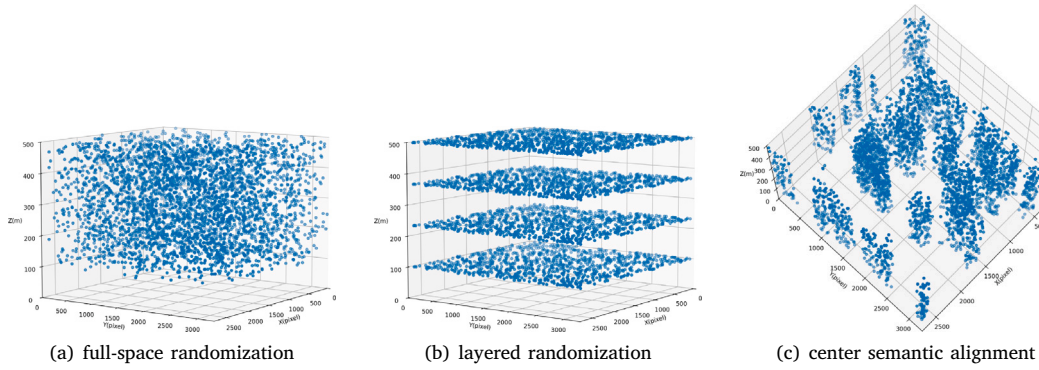


Fig. 17. Comparison of three particle initialization strategies.

altitudes and the true UAV altitude significantly degrade particle fitting and convergence performance.

5.2.4. Effect of the center distance field

To further enhance alignment, a center distance field is incorporated into the particle weight formulation. Under identical experimental settings on MAFS-10, introducing the center distance field improves the average particle fitting rate by 5.8% compared with the baseline, as visualized in Fig. 18. In addition, the post-fitting particle distributions exhibit reduced scattering, indicating more concentrated and stable convergence.

5.2.5. Semantic weight allocation

We analyze the weight distribution for each semantic category. Localization methods based on a single semantic fail consistently. When testing localization methods that combine two semantics (e.g., road category with any other category), all combinations fail due to insufficient feature information, particularly because most roads in the test scenarios are located around the perimeter. In contrast, all combinations of the three semantics complete the fitting, though at a slower rate than when all four semantics are employed simultaneously.

5.3. Comparison with the existing state-of-the-art

Section 2 discusses satellite imagery-based UAV localization systems most comparable to our approach, including works by Masselli et al. [21], Mantelli et al. [8], Wu et al. [6], Dai et al. [13], and Hong et al. [41]. Table 6 provides a systematic comparison of methodological parameters, with limited analysis due to unavailable datasets or codes from benchmark studies.

A critical evaluation metric for satellite-based localization is the image-to-ground scale factor (pixel-to-meter resolution ratio). Larger scale factors increase per-pixel distance representation, amplifying positioning errors and algorithmic precision requirements. Our dataset exhibits superior scale characteristics, being 6.02× smaller than [21] and 1.08× smaller than [8].

Our system achieves positioning errors under 9 m while overcoming critical limitations of existing approaches. Unlike the deep learning-based method [13], which requires scene-specific pretraining and constrained search spaces, we enable localization across a 1000× larger 3D search volume. While matching the accuracy of Hong et al. [41] and Wu et al. [6], our approach reduces prior information requirements to a single low-resolution 2D satellite image, contrasting with UAVD4L's computationally intensive 6-DoF localization framework that demands detailed 3D textured models.

We further evaluate performance through the normalized error-to-map ratio (EMR: average error/map dimension). Lower EMR values indicate superior scalability, with our system achieving a benchmark 4.5 % ratio, which is 3.4× and 17.7× lower than [8,21], respectively.

Furthermore, compared to end-to-end deep learning localization approaches, SWA-PF offers distinct characteristics regarding interpretability and fault diagnosis. End-to-end neural networks typically function by directly regressing pose estimates from input image pairs without explicit intermediate representations. Consequently, when localization errors occur, attributing the failure to specific components – such as feature extraction, cross-view matching, or pose regression – can be challenging. In contrast, the SWA-PF framework decouples the localization process into distinct functional stages: semantic segmentation, kinematic state prediction, and measurement updating. This modular structure allows for the inspection of intermediate outputs, such as the Semantic-Weighted Distance Maps (SWDM), and the observation of

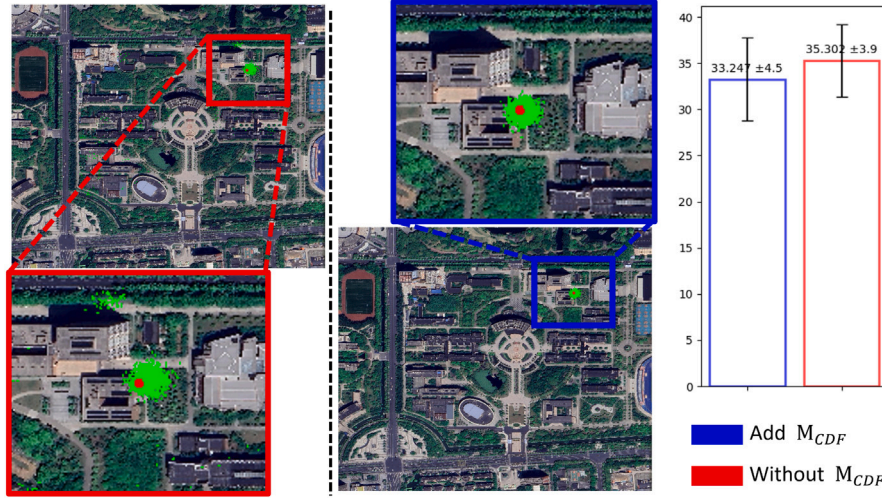


Fig. 18. Comparison of performance and fitting rate with MCDF. Particle state diagrams are recorded with and without the M_{CDF} method, showing improved fitting rate and central error after incorporating the M_{CDF} method.

Table 6
Comparison with existing state-of-the-art methods.

Method	Map size (m ²)	Map size (p ²)	Trajectory (m)	Error (m)	Error (p)
UAVD4L [6]	2425 × 1054	–	–	<5	–
abBRIEF [8]	1160 × 1160	4800 × 4800	2400	17.78	74.08
MasselliUL14 et al. [21]	150 × 90	3440 × 3440	<500	12.75	63.7
Hong et al. [41]	–	–	600	<10	–
SWA-PF (ours)	2000 × 2000	7615 × 5258	4000	<9	<35

Table 7
Summary of MAFS dataset and SWA-PF localization performance.

Dataset Statistics (MAFS/SemanticMAFS)	
Geographic Coverage	14 campuses + 1 urban region (Hangzhou, China)
Total Trajectory Length	40.5 km
Altitude Range	100 m–500 m
Largest Map Size	2 km × 2 km (7615 × 5258 pixels)
Flight Type	Fixed-altitude + Variable-altitude
UAV Semantic Classes	7 (roof, wall, ground, road, vegetation, water, vehicle)
Satellite Semantic Classes	4 (building, road, vegetation, water)
Main Localization Performance (SWA-PF)	
Localization DoF	4-DoF (x, y, h, θ)
RMSE (MAFS-10)	23.50 p
Median Error	19.02 p
Recall@10 m	95.35%
Convergence Time	≈ 9 s
Total Inference Time	31 s (MAFS-10)
Particle Number	5000 (fixed)/40,000 (variable altitude)
GPU Memory Peak	3499.58 MB

particle convergence behavior. As a result, positioning anomalies can be more directly traced to specific environmental ambiguities or motion model deviations, providing an analytical capability that is relevant for safety-critical UAV applications.

5.4. Compact overview of dataset and experimental performance

To provide readers with a concise overview of both dataset characteristics and the main performance outcomes of the proposed SWA-PF framework, we summarize the key statistics and experimental results in Table 7. This compact table facilitates quick reference and highlights the scale and effectiveness of the proposed system.

5.5. Summary of experiments

We conducted extensive experiments on the MAFS dataset under fixed-altitude and variable-altitude conditions. The proposed SWA-PF

achieved approximately a tenfold speedup over classical feature-based methods within the same particle filter framework. The global positioning RMSE remained below 25 p, and the recall for errors under 10 m exceeded 95% on the MAFS-10 trajectory. Full 4-DoF pose estimates, including altitude and yaw, were obtained within seconds using only a low-resolution satellite map. SWA-PF also demonstrated strong robustness to map resolution degradation and altitude variations. Statistical analysis confirmed significant improvements over all baselines ($p < 0.05$) with large effect sizes ($|\delta| \geq 0.474$). These results show that SWA-PF effectively balances localization accuracy, computational efficiency, and memory usage for onboard UAV deployment.

6. Conclusion

This work tackled key limitations of retrieval-based UAV localization by introducing the multi-altitude MAFS dataset with pixel-level

semantic annotations (SemanticMAFS) and proposing the Semantic-Weighted Adaptive Particle Filter (SWA-PF). SWA-PF combines a compact 4-DoF particle representation of strictly linear memory complexity, a semantic weighting mechanism that leverages structurally stable scene layouts, and an optimized adaptive filter. These designs yield a tenfold end-to-end speedup over conventional feature extraction within the same particle filter framework, keep the global positioning error below 10 m relative to onboard GNSS, and deliver 4-DoF pose estimates within seconds. Crucially, SWA-PF shows exceptional robustness to low-resolution satellite imagery: unlike appearance-driven methods, it relies on high-level structural distributions rather than fine textures, sustaining accuracy even when map resolution is halved. This enables large-scale UAV localization with lightweight, storage-efficient maps, eliminating the need for high-resolution data in bandwidth-limited or mission-critical settings.

7. Limitations

Despite these advances, SWA-PF exhibits several limitations. First, severe class imbalance – particularly when vegetation exceeds 50% – flattens the posterior distribution and slows convergence. Second, extreme domain shifts such as heavy snow or highly unstructured rural terrains violate the urban structural priors, causing matching failures. Third, the current implementation depends on GPU acceleration; CPU-only execution of SegFormer inference and matrix operations is approximately 15× slower, challenging real-time deployment on resource-constrained platforms.

8. Future work

Building on these findings, future research will proceed along three main directions. We will develop domain-adaptive semantic weighting schemes to extend robustness to rural and unstructured environments, and design uncertainty-aware mechanisms that mitigate segmentation noise. Furthermore, we aim to reduce reliance on continuous IMU input by integrating lightweight global re-localization modules, enabling the system to recover from tracking loss and operate reliably in intermittent scenarios.

CRedit authorship contribution statement

Jiayu Yuan: Writing – original draft, Software, Resources, Formal analysis, Data curation. **Ming Dai:** Writing – review & editing, Writing – original draft, Supervision, Data curation. **Enhui Zheng:** Validation, Supervision, Project administration, Investigation, Funding acquisition. **Chao Su:** Validation, Software, Investigation, Data curation. **Nanxing Chen:** Validation, Software, Investigation, Data curation. **Qiming Hu:** Validation, Software, Investigation, Data curation. **Shibo Zhu:** Validation, Software, Investigation, Data curation. **Yibin Cao:** Validation, Software, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patcog.2026.114522>.

Data availability

Data will be made available on request.

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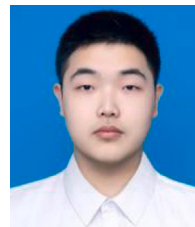
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